Constrained Extrema: Tangent Spaces, Second Variation, and Dual Variables

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At a constrained extremum, one is free to move along the feasible set but not away from it. Lagrange multipliers express the vanishing of the objective's first-order change along every allowed direction. The Hessian test must then measure curvature along those same directions, including the curvature of the constraint itself. This leads naturally to bordered Hessians and to the role of multipliers as sensitivities in convex optimization.[1]

§1.1 First variation is normal to the feasible surface

Let $f, g : \mathbb{R}^n \to \mathbb{R}$ be C^2, and suppose $g(x_*) = 0$ with $\nabla g(x_*) \neq 0$. The implicit function theorem says that the feasible set near x_* is a smooth hypersurface, with tangent space

$$T_{x_*}M = \{v \in \mathbb{R}^n : \nabla g(x_*) \cdot v = 0\}.$$

If $f|_M$ has a local extremum at x_*, then $Df(x_*)v = 0$ for every $v \in T_{x_*}M$. The annihilator of this tangent hyperplane is the line spanned by $Dg(x_*)$. Hence

$$\boxed{\nabla f(x_*) = \lambda \nabla g(x_*)}$$

for some scalar λ. The multiplier is the coefficient expressing the objective's differential in the normal direction.

For several constraints $g = (g_1, \dots, g_m)$, the same argument works if the rows of $Dg(x_*)$ are independent:

$$\nabla f(x_*) = Dg(x_*)^T \lambda.$$

Independence is a constraint qualification, ensuring that every vector satisfying the linearized constraints is tangent to an actual feasible motion.

Example 1.1.1 (Finding the nearest point is already a multiplier calculation)
Find the point on $x + y = 1$ nearest the origin. Minimize $f(x, y) = x^2 + y^2$ subject to $g(x, y) = x + y - 1 = 0$. The multiplier equations are

$$(2x, 2y) = \lambda(1, 1), \qquad x + y = 1,$$

giving $\boxed{x = y = \frac{1}{2}}$ and $\lambda = 1$. Substitution $y = 1 - x$ independently verifies that

$$x^2 + (1 - x)^2 = 2\left(x - \frac{1}{2}\right)^2 + \frac{1}{2}$$

has its unique global minimum there. The tangent vector $(1, -1)$ gives a positive restricted quadratic form:

$$(1, -1)H_f(1, -1)^T = 4 > 0.$$

[1] For constrained second-order conditions and multiplier sensitivity, see Boyd and Vandenberghe, *Convex Optimization*, Chapters 5 and 10: https://web.stanford.edu/~boyd/cvxbook/.

For an affine constraint $H_g = 0$, so here the Lagrangian Hessian agrees with H_f.

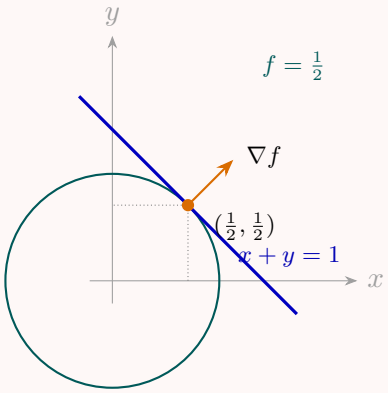


Figure 1.1: The smallest squared-distance level set touching $x + y = 1$ has its normal aligned with the constraint normal.

§1.2 Why the ordinary Hessian is not the right one

Choose the Lagrangian $L(x, \lambda) = f(x) - \lambda g(x)$, and let $x(t)$ be any twice-differentiable feasible curve with $x(0) = x_*$, $x'(0) = v \in T_{x_*}M$. Differentiating $g(x(t)) = 0$ twice gives

$$Dg(x_*)x''(0) + D^2g(x_*)[v, v] = 0.$$

Meanwhile,

$$\frac{d^2}{dt^2} f(x(t))\Big|_{t=0} = D^2 f(x_*)[v, v] + Df(x_*)x''(0).$$

At a stationary point $Df = \lambda Dg$, so the second equation becomes

$$\boxed{\frac{d^2}{dt^2} f(x(t))\Big|_{t=0} = v^T \big(H_f(x_*) - \lambda H_g(x_*)\big)v.}$$

The curvature of the constraint has cancelled the acceleration of the chosen path. This is the *restricted Hessian of the Lagrangian*. For a strict constrained minimum, it suffices that the quadratic form be positive definite on nonzero tangent vectors; negative definiteness similarly gives a strict local maximum. Indefiniteness on the tangent space rules out either kind of extremum.

§1.3 The sphere reveals the correction term

Maximize or minimize $f(x, y) = x$ on the unit circle $g(x, y) = x^2 + y^2 - 1 = 0$. The stationarity equations are

$$(1, 0) = \lambda(2x, 2y), \qquad x^2 + y^2 = 1,$$

giving the two points $p_+ = (1, 0)$, $p_- = (-1, 0)$, with $\lambda = 1/2$ and $-1/2$ respectively.

Since $H_f = 0$ and $H_g = 2I$, the Lagrangian Hessian is

$$H_L = -2\lambda I.$$

At either point the tangent space is the vertical line, spanned by $v = (0, 1)$. At p_+, $v^T H_L v = -1 < 0$, so the point is a strict constrained maximum. At p_-, the value is $+1 > 0$, a strict constrained minimum.

The objective is linear and its ordinary Hessian vanishes identically. All the relevant second-order behavior comes from the curvature of the circle. This is why checking H_f alone would miss the entire problem.

§1.4 Bordered Hessians encode the same restriction

In two variables with one regular constraint, write $J = (g_x, g_y)$ and let $H = H_L$. A nonzero tangent vector is $v = (-g_y, g_x)$. The bordered matrix is

$$B = \begin{pmatrix} 0 & g_x & g_y \\ g_x & H_{11} & H_{12} \\ g_y & H_{12} & H_{22} \end{pmatrix}.$$

Direct expansion shows

$$\boxed{\det B = -v^T H v.}$$

Thus, for this particular dimension and convention, a negative bordered determinant corresponds to positive constrained second variation, while a positive determinant corresponds to negative second variation. In higher dimensions the signs and lists of principal minors depend on the number of constraints; the invariant statement is always definiteness of H_L on $\ker Dg$. This avoids memorizing ambiguous bordered-Hessian sign tables.

§1.5 Singular constraints change the rules

Take the cusp

$$g(x, y) = y^2 - x^3 = 0$$

and the objective $f(x, y) = x$. Every feasible point satisfies $x \geq 0$, so the origin is a strict global minimum. Yet

$$\nabla g(0, 0) = 0, \qquad \nabla f(0, 0) = (1, 0),$$

and no multiplier can satisfy $\nabla f = \lambda \nabla g$ there. The feasible set is singular at the origin, so the regular Lagrange theorem does not apply.

Even with a regular constraint, a vanishing restricted Hessian does not determine the type of stationary point. For $f(x, y) = x^4 - y^4$ on the whole plane, the gradient and Hessian vanish at the origin, but positive and negative values approach along the coordinate axes. Higher-order terms resolve the degeneracy.

§1.6 Multipliers also measure the price of a constraint

Consider a convex differentiable $f : \mathbb{R}^n \to \mathbb{R}$ minimized under an affine constraint

$$Ax = b.$$

Use $L(x, \lambda) = f(x) + \lambda^T (Ax - b)$. If x_* is feasible and $\nabla f(x_*) + A^T \lambda_* = 0$, convexity implies, for any feasible x,

$$f(x) \geq f(x_*) + \nabla f(x_*)^T (x - x_*) = f(x_*) - \lambda_*^T A(x - x_*) = f(x_*).$$

Thus the multiplier stationarity condition is globally sufficient in this convex setting.

Let $v(b)$ denote the optimal value when the constraint is $Ax = b$, and assume locally that a smooth optimum and multiplier exist. Differentiating the optimal-value equation gives

$$dv(b) = -\lambda_*^T \, db.$$

A multiplier is therefore a shadow price: it measures the first-order change of the optimal objective when the permitted constraint value moves. In the circle example the multiplier encodes normality; in convex optimization the same normal coefficient becomes a quantitative sensitivity.

§1.7 When a boundary pushes back: an inequality example

The nearest-point problem changes subtly if we replace $x + y = 1$ by the half-plane $x + y \geq 1$. Minimize $x^2 + y^2$ on this set. Define

$$h(x, y) = 1 - x - y \leq 0, \qquad L(x, y, \mu) = x^2 + y^2 + \mu(1 - x - y).$$

For this convex problem the Karush–Kuhn–Tucker conditions read

$$\begin{cases} 2x - \mu = 0, & 2y - \mu = 0, \\ x + y \geq 1, & \mu \geq 0, \\ \mu(1 - x - y) = 0. \end{cases}$$

The last condition is complementary slackness: an inactive inequality has multiplier zero, while a positive multiplier requires equality at the boundary. If $\mu = 0$, then $x = y = 0$, which is infeasible. Thus the constraint is active and

$$\boxed{(x_*, y_*) = (\tfrac{1}{2}, \tfrac{1}{2}), \qquad \mu_* = 1.}$$

Strict convexity and feasibility certify the global minimum. The multiplier's nonnegative sign has a geometric meaning: the active boundary can only push toward the feasible side.

A warning about numerical methods emerges from this example. Squaring a constraint expression h as ρh^2 punishes even strictly feasible points with $h < 0$. A quadratic penalty for $h \leq 0$ should instead use $\rho[h]_+^2$ with $[h]_+ = \max(h, 0)$. Even then a finite quadratic penalty generally does not enforce exact feasibility. An interior-point method takes a different route, adding a barrier $-\tau \log(-h)$ only where $h < 0$. These constructions approximate or enforce the feasible boundary in genuinely different ways.

§1.8 Second-order silence and global existence

A zero or semidefinite restricted Hessian is a statement that the second-order test is inconclusive. At the origin,

$$f_+(x, y) = x^4 + y^4, \qquad f_-(x, y) = x^4 - y^4$$

have identical zero gradients and Hessians, but f_+ has a strict minimum and f_- a saddle. The same contrast can be realized on a regular two-dimensional constraint manifold by adding an extraneous coordinate z with constraint $z = 0$. Higher-order terms matter when the quadratic approximation cannot distinguish signs.

Existence of global extrema asks a logically separate question. The function x has no extremum on the open interval $(0, 1)$; there is no stationary point to find. On a compact feasible set, however, every continuous objective attains minimum and maximum. The circle in the earlier example is compact, so comparing its two stationary values actually settles the global question.

A deeper connection is Morse theory: at a nondegenerate critical point on a smooth manifold, the number of negative eigenvalues of the restricted Hessian is its Morse index. The very quadratic form used to test constrained extrema also gives local information from which changes in the manifold's topology can be studied. The difficult global theorems lie beyond this example, but the local datum entering them is explicit.

The tangent space tells us which infinitesimal motions are allowed, the restricted Hessian classifies their second-order cost, and the multiplier measures the cost of changing the constraint itself. These three views belong to one differential-geometric calculation.